

Exploring Gesture and Prosodic Alignment at Code-Switching Boundaries: A Multimodal Analysis of Twi-English Speech

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Abstract

This study investigates gesture-prosody coordination in naturalistic Twi-English code-switching speech, focusing on gesture type distribution, temporal alignment between gesture strokes and prosodic prominence, and gestural activity at code-switching boundaries. A corpus of 11 video clips totalling 65.1 minutes was annotated using ELAN 7.0, yielding 837 gesture strokes classified according to McNeill's typology. Fundamental frequency was extracted using Praat's autocorrelation algorithm, and code-switching points were identified through a computational language inference pipeline built on FFmpeg speech detection. Results show that beat gestures dominated the corpus at 68.0%, consistent with the prosodic-structuring demands of the speech genres studied. Gesture stroke onsets clustered near local pitch peaks, with a mean signed time difference of -36.8ms and a median of -14.0ms. 51.6% of strokes fell within 50ms of a pitch peak and 70.8% within 100ms, with broadly similar patterns across gesture types. Gesture density was 2.30 times higher within ± 500 ms of code-switching points than in surrounding speech, a difference confirmed as statistically significant at $t(10) = 6.51$, $p < .001$. These findings demonstrate that code-switching in Twi-English bilingual discourse is a multimodal event in which gesture and prosody jointly mark language transitions. The study contributes an empirical account of bilingual multimodal communication in an understudied African language contact setting and presents a fully reproducible computational analysis pipeline for multimodal bilingual corpus research.

Introduction

Human communication is far richer than the words we choose to speak. Alongside language, the voice itself carries meaning through variations in pitch, rhythm, and volume while the body contributes through gesture, facial expression, and posture. These channels do not operate in isolation. Prosody and gesture in particular work together as interconnected systems to construct meaning in natural interaction. A number of recent studies in multimodal communication have shown that modalities do not operate in parallel but rather are complementary in ways that no single modality could manage (Brown & Prieto, 2021; Holler & Levinson, 2019). It has been documented in detail that gestures and intonation co-produce prominence in a range of contexts (Brown & Prieto, 2021). Holler and Levinson (2019) advance a unified theory of multimodal communication in which gesture is constitutive rather than supplementary. Both positions are influenced by a common conception of the relation of speech and gesture from McNeill's (2005) growth point perspective. This view holds that speech and gesture do not run as two parallel systems but emerge from a single cognitive

impulse, and come together at the prosodically most prominent syllable of speech. In this way, speech and gesture co-express the same underlying communicative moment (McNeill & Duncan, 2000, p. 141). This view of gesture and speech relationship shapes the analytic approach in this study.

Most of this evidence comes from the literature on monolingual speech. In this literature, gestures have been found to be deeply intertwined with prosodic events such as pitch peaks, stress and rhythmic beats. This suggests that the acoustic and kinesic channels are tightly connected both in time and meaning (McNeill, 2005; Jiménez-Bravo & Marrero-Aguiar, 2024). Gestural strokes are often timed with prosodically prominent syllables, supporting the idea that gesture and prosody are co-expressive systems (Brown & Prieto, 2021). Code-switching in particular provides additional prosodic dimensions that monolingual phonetic models would struggle to account for. The switch between languages is measurably accompanied by a pitch reset and a change in pitch range, with bilinguals often raising pitch when switching languages (Zeng, 2024). Research has documented that bilinguals actually make deliberate adjustments to how they sound at the switching point. They often place their two languages in separate intonation units in order to separate them and avoid structural incompatibilities (Torres Cacoullos, 2020). Similarly, in the gestural side, the timing relationship between gestures and speech stays basically intact even in bilinguals and there is no major breakdown in synchrony just because someone speaks two languages (Graziano, Nicoladis & Marentette, 2020). However, it remains to be tested whether synchrony is also preserved at the exact moment of code-switching where prosodic recalibration is most active.

The connection between gesture and speech at the neurophysiological and biomechanical level has been observed to be partly physical. The efforts involved in the production of gestures have been found to directly affect how the voice sounds at that moment, including how high or low the pitch is (Pouw, Harrison and Dixon, 2019). This proposal has been proven to be consistent and repeatable. In corpus-based studies, the peak of a gesture and the peak of pitch in speech land almost at the exact same moment, within milliseconds of each other in both casual conversations and formal speech (Loehr, 2012; Pouw and Dixon, 2019). Nonetheless, there is a gap in the literature of bilingual multimodal communication. In a thorough review of gesture and bilingualism, Gullberg (2013) explains how little research has been conducted on gestures in adult code-switching (p. 428) and further emphasizes that the correlation between phonology, prosody and gesture among bilinguals is a field that is yet to be systematically studied (p. 431). In Africa, the role of gesture in signalling or accompanying code-switching transitions, especially in countries such as Ghana, has received little or no systematic attention. The present study responds to that gap.

This gap is particularly critical in the Ghanaian bilingual situation. Twi is a Niger-Congo language belonging to the Akan of the Kwa branch. It is a tonal language with the lexical meaning of a word being determined by pitch as opposed to pitch indicating sentence-level prosodic structure. This is different from how pitch works in English, which is a stress-timed language where intonation and intonational prominence are indicated by a combination of pitch, duration and intensity to cue information structure. Both systems are described in detail

by Oppong (2024) as he demonstrates that Akan bases its system on intensity and duration as important prosodic indicators in addition to pitch and that its prosodic behaviour is noticeably different from European or North American varieties. Ghanaian English has its own unique prosodic features and as such, it is not comparable to the varieties on which most multimodal studies are based. Due to the radically different pitch organization of the two languages, pitch peaks in Twi-English speech cannot be interpreted as an equivalence of the prosodic prominence peaks that the studies of monolingual gesture-prosody alignment tend to presume. Gullberg (2013) here arrives at an equally relevant point that gesture timing between bilinguals does not depend on language, with Spanish-English bilinguals aligning path gestures with verbs in English in patterns that are determined by Spanish conceptual organization and not English monolingual conventions, because the underlying language active at the moment of production influences which gestural timing patterns emerge (pp. 426-427). A tonal language environment like Twi-English where pitch functions as a lexical rather than intonational cue, represents a fundamentally different condition from the bilingual settings studied so far. This warrants an investigation of Twi-English on its own terms rather than through extension of existing frameworks. In this study, the corpus examined represents Twi as the matrix language, the one carrying the structure of the discourse and the English language as the embedded language that is inserted into Twi-dominated stretches at both lexical and phrasal boundaries (Myers-Scotton, 1993). Such a structural asymmetry is central to the design of this study and is presented through the annotation strategy outlined in section 2.3. Beyond the linguistic differences, another dimension of non-conformity to the research norms in the West is the cultural practices of gesture in Ghanaian speech communities. Kita and Essegbey (2001) reported this with respect to pointing, having established that culturally based limitations on the use of the left-hand result in gestural behaviours that are different from those reported in many European contexts. Gestural repertoires in Ghana are shaped by local norms, and accounting for them is deemed necessary. Twi and English are freely and commonly used and practised throughout the Ghanaian social life, but little is known about the multimodal relations of this contact environment.

The body of research that directly analyses gesture in code-switching contexts was already described as very small in the field's most comprehensive review to date (Gullberg, 2013, p. 428). The intervening years have done little to change that. The first systematic investigation is a study conducted by Raffler-Engel (1976) in Canada on bilingual children which found that French speaking children would modify their gestural behaviour according to the ethnicity of the addressee whereas English-speaking children preserved their gestural behaviour irrespective of the language. Despite its methodological shortcomings, that study revealed the possibility that gestures are not merely following language switches the way lexical choices do. The closest related study that was done on adults is that of Lacroix and Rioux (1978) who observed Canadian English- French bilingual groups. They observed that there was not much indication of gestural behavioural variation when the language was swapped between speakers. Most importantly, though outside observers could still determine the dominant languages of speakers using visual input only. This indicates that there are still underlying latent differences between gestural repertoires which happen even in the absence of overt switching. At a community level, as speakers gradually shifts from one language to another across generations,

gestures eventually follow but they take longer to catch up. There is a period where gesture forms from both languages get mixed together and there is no real-time correlation between a spoken switch and a simultaneous gesture switch. All these studies have examined whether bilinguals use different gesture types or styles in their two languages. None of them questions whether there is a change in density of gestures or a temporal alignment during the time of a code-switch. It is a different question from what those earlier studies were looking at and this is the question that this study seeks to answer. Rather than examining gestural form, the present study asks whether the quantitative and temporal properties of co-speech gesture are sensitive to the structural event of a language transition regardless of what kind of gesture is produced. Although Wu (1996) explored the relationship between gesture and code-switching through growth point theory, no prior study has tested how gesture density or prosodic alignment is disrupted at the switching boundary itself, let alone in a setting involving a tonal African language.

The speakers of this corpus are adults who use English as a second language, acquired in a Twi dominant environment. Gullberg's (2013) review of adult bilingual gesture use which covers proficiency effects, compensation and cross-linguistic influence gives the broader scholarly context for this population. She records that it is difficult to give simple compensatory accounts because adult bilinguals have more gestures in the weaker language than in the stronger language but iconic gestures are more frequent in the stronger language, (pp. 424-425). What matters most for this study specifically is her observation that pragmatic and the structure of discourse gestures are among the most common in bilingual adult speech, as they occur at all places wherever the demands of managing complex, fluent speech require manual rhythmic support (p. 425). The kind of adult bilingual speakers that Gullberg is reviewing are the speakers who work in the highly competitive and rhetorically demanding genres that are represented in this corpus, including political commentary and motivational address.

This study aims at achieving three goals using naturalistic Twi-English bilingual speech:

1. To analyze the distribution of different gesture types in Twi-English code-switching discourse.
2. To examine the alignment between gesture strokes and prosodic prominence in Twi-English code-switching discourse.
3. To assess whether gesture frequency and prosodic prominence increase at code-switching points in Twi-English discourse.

The analysis is based on gesture annotation, acoustic pitch extraction and statistical modelling to delineate the real-time coordination of the visual and acoustic channels. This study is positioned as a cross-disciplinary study of linguistics, cognitive science and data-driven multimodal analysis to contribute not only to theoretical knowledge about cross-modal coordination but also to an empirical history of bilingual communication within an African language contact environment that has so far received very little systematic investigation.

2. Methodology

This study adopts a mixed method approach, specifically a multimodal corpus based method, to analyse gesture prosody coordination in Twi English speech. This section describes the corpus, annotation procedures, acoustic analysis approach and the computational pipeline developed to identify code-switching points and measure gesture and prosody coordination.

2.1 Data and Corpus Construction

This study draws on naturalistic Twi English bilingual speech which was retrieved from YouTube. A corpus-based approach was chosen over experimental speech because naturalistic data reveals the spontaneous code-switching that is often suppressed in laboratory tasks (Torres Cacoullos, 2020). In controlled settings, speakers tend to reduce the rate of switching and change their prosodic and gesture patterns when they realise that they are being observed. This is why naturally occurring video data is appropriate for this study. The annotated corpus files, data outputs and the scripts for analysis used in this study are publicly archived and freely accessible (Aryee, 2026).

The four source videos were selected based on four criteria: (1) frequent and natural Twi English code switching, (2) clear visibility of the speaker’s hands, (3) audio quality sufficient for pitch extraction, and (4) predominantly monologic speech with minimal interlocutor overlap. Monologic segments were chosen because they provide longer and uninterrupted stretches of rich gesture speech that are easier to annotate and examine systematically than highly interactive dialogue. The videos are drawn from two settings. These consist of motivational public addresses and political talk show segments. These genres elicit expressive speech with frequent gesture use, which is relevant to this study. From these four videos, eleven clips were extracted, yielding a total of approximately 65.1 minutes of annotated speech, as described in Table 1. Each clip was assigned a unique identifier, GH001 through GH011 to facilitate cross referencing across annotation and analysis

The corpus includes four Ghanaian public figures. Abronye DC, a politician and media personality, appears in clips GH001 through GH003. Lawyer Anokye Frimpong, a legal and political commentator, appears in clips GH004 through GH007. Dr. Beatrice Nana Ofoah, a motivational speaker, contributes clips GH008 and GH009. Okatakyie Afrifa, also a political commentator, appears in clips GH010 and GH011. Three of the four speakers are male and one is female. All are adults who speak Akan/Twi as their first language and acquired English through the Ghanaian educational system, fitting the bilingual profile outlined in the introduction.

Table 1. Corpus Summary

Clip	Speaker	gender	Genre	Duration (min)	Gestures	Switch Points
GH001	Abronye DC	Male	Political talk show	5.6	46	114
GH002	Abronye DC	Male	Political talk show	2.5	44	62
GH003	Abronye DC	Male	Political talk show	6.9	70	132

Clip	Speaker	gender	Genre	Duration (min)	Gestures	Switch Points
GH004	Lawyer Anokye Frimpong	Male	Political discussion	4.0	68	52
GH005	Lawyer Anokye Frimpong	Male	Political discussion	2.1	31	46
GH006	Lawyer Anokye Frimpong	Male	Political discussion	2.9	33	78
GH007	Lawyer Anokye Frimpong	Male	Political discussion	4.8	82	68
GH008	Dr. Beatrice Nana Ofosuah	Female	Motivational address	10.0	133	164
GH009	Dr. Beatrice Nana Ofosuah	Female	Motivational address	10.0	119	81
GH010	Okatakyie Afrifa	Male	Political commentary	8.6	136	153
GH011	Okatakyie Afrifa	Male	Political commentary	7.8	75	139
TOTAL				65.1	837	1,089

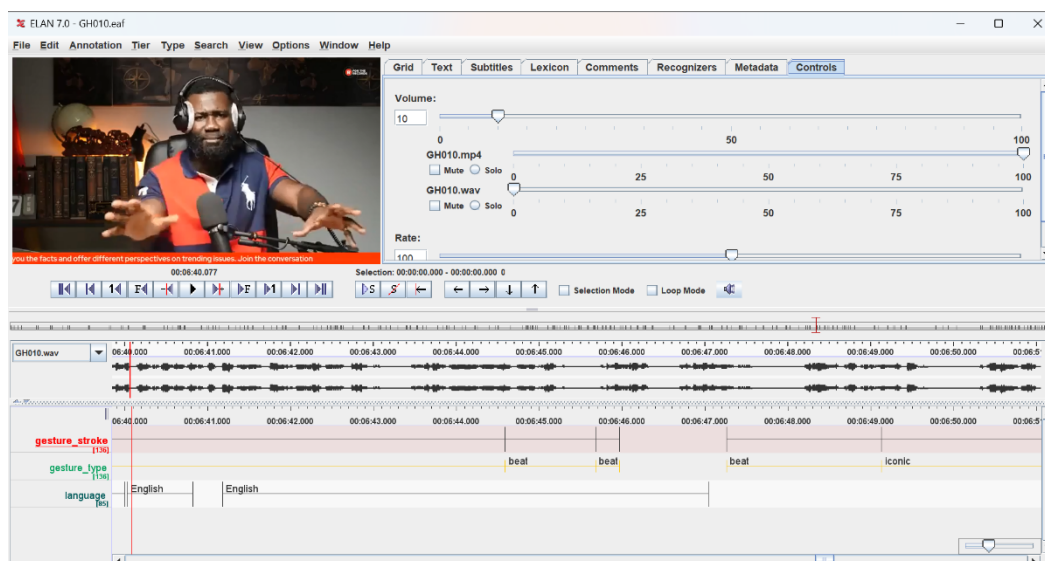
Note. G/s = gesture strokes per second. Switch points identified at all Twi-English language boundaries within detected speech segments.

2.2 Gesture Annotation

Gesture annotation was carried out in ELAN version 7.0, a widely used annotation tool for video and audio recordings (Wittenburg et al., 2006). Three tiers were created for annotation. The first was the gesture stroke tier which was set as an alignable parent tier. It was used to mark the onset and offset of each gesture stroke. The stroke phase was chosen as the main unit of analysis because it is the peak phase of a gesture and is typically the phase most closely synchronised with prosodic prominence (McNeill, 2005; Jimenez-Bravo and Marrero-Aguilar, 2024). Recording stroke boundaries in lieu of the entire gesture phrase made it possible to match stroke timings to acoustic measures downstream with high accuracy. The second tier which is gesture type was set up as a dependent child tier of gesture stroke and it was used to categorize each stroke according to McNeill's (2005) gesture typology.

In gesture type, three categories were coded. The first one was beat gestures, which are rhythmic movements that mark prosodic or discourse structure without carrying representational content as described by Gullberg (1998, p. 51) as “simple gestures, not depicting anything, but aligned with prosodic prominence patterns in speech” that “keep the same form regardless of content”. The second type was iconic gestures. These bear a formal resemblance to the semantic content they accompany. The third type was deictic gestures, which perform a pointing function involving movements that direct attention to something either physically present or being referenced in speech. This three category scheme was chosen because it covers the functional range of co speech gestures that is likely to interact with prosodic structure in this corpus.

Figure 1. ELAN 7.0 annotation interface showing a segment from clip GH010.



A total of 837 gesture strokes were annotated across the eleven clips. Annotation was conducted by a single trained coder. While inter-rater reliability assessment would strengthen the coding scheme, single-annotator coding is common in exploratory multimodal corpus studies where the primary aim is pattern detection rather than fine-grained perceptual judgment (Loehr, 2012). This is acknowledged as a limitation and is discussed further in the conclusion.

2.3 Language Segmentation and Switch Point Identification

Identifying code-switching points required constructing a full language timeline for each clip. English intervals were annotated directly in ELAN as the third tier, Language, an independent alignable tier applied whenever the speaker transitioned into English. Twi was not annotated as a separate label because it functioned as the matrix language throughout, meaning the default medium of speech was Twi and English represented the embedded language. This annotation strategy is consistent with established practice in matrix language frame approaches to bilingual corpus construction (Myers-Scotton, 1993).

To infer Twi intervals computationally, speech activity detection was first applied to each clip using FFmpeg's silence detect filter, with a noise threshold of -35 dB and a minimum silence duration of 300ms. These parameters were selected to balance sensitivity to genuine speech boundaries against over-segmentation caused by brief pauses and breath groups. Speech segments were then exported, and Twi intervals were derived by subtracting the annotated English intervals from the total detected speech time. Any residual speech not labelled as English was treated as Twi.

This inference-based approach is efficient and reproducible, but it carries an important limitation. Non-linguistic vocalizations such as laughter, filled pauses, and audible breath are not separately coded and are therefore absorbed into the Twi category. This may slightly inflate Twi interval counts and introduce noise into switch point detection. The effect is difficult to quantify without exhaustive manual coding of every non-linguistic event, which falls outside

the scope of this study. It is acknowledged here as a methodological constraint that future work should address through explicit vocalization coding.

Code-switching points were identified at every boundary where the inferred language label changed from Twi to English or English to Twi. A total of 1,089 switch points were detected across the corpus, split almost equally between English-to-Twi transitions ($n = 546$, 50.1%) and Twi-to-English transitions ($n = 543$, 49.9%). To define the temporal zone of interest around each switch, a window of $\pm 500\text{ms}$ was applied. This 500ms window was chosen because it spans the approximate duration of one to two syllables in conversational speech, covering the prosodic and gestural activity most likely to be associated with the transition point (Zeng, 2024).

2.4 Acoustic Prosodic Analysis

Fundamental frequency (F0) was extracted as the primary acoustic correlate of prosodic prominence. F0 is well established as a central cue to pitch accent, stress, and information-structural salience in both tone and intonation languages, making it the appropriate measure for this analysis (Oppong, 2024). Extraction was performed using Praat's autocorrelation-based pitch tracking algorithm, To Pitch (ac), which is the standard method in speech research for its robustness to voicing irregularities and its suitability for naturalistic, non-read speech (Boersma & Weenink, 2024).

Because Praat processes audio more reliably from WAV format than from compressed video, each MP4 clip was first converted to a mono 16kHz WAV file using FFmpeg prior to analysis. An F0 range of 75 to 300 Hz was specified, covering the expected pitch range for adult Ghanaian male and female speakers. A time step of 10ms was used to produce a temporally dense F0 contour suitable for alignment with gesture annotation timestamps. Frames where Praat returned no voiced pitch estimate were excluded, retaining only frames with confident F0 values. From the resulting contour, local F0 maxima were identified as pitch peaks, yielding 28,668 peaks across the corpus. All extracted values fell within the specified 75 to 300 Hz range, confirming that the extraction captured genuine speech F0 rather than harmonics or noise artifacts.

2.5 Gesture-Prosody Alignment

To examine the temporal relationship between gesture and prosodic prominence, each gesture stroke onset was matched to the nearest pitch peak within the same clip. The time difference was computed as gesture onset minus pitch peak time in milliseconds, such that negative values indicate the gesture stroke precedes the pitch peak and positive values indicate it follows. This nearest-peak matching approach was adopted because it is computationally tractable at corpus scale and does not require phonemic transcription or forced alignment, making it well suited to an exploratory study of this kind.

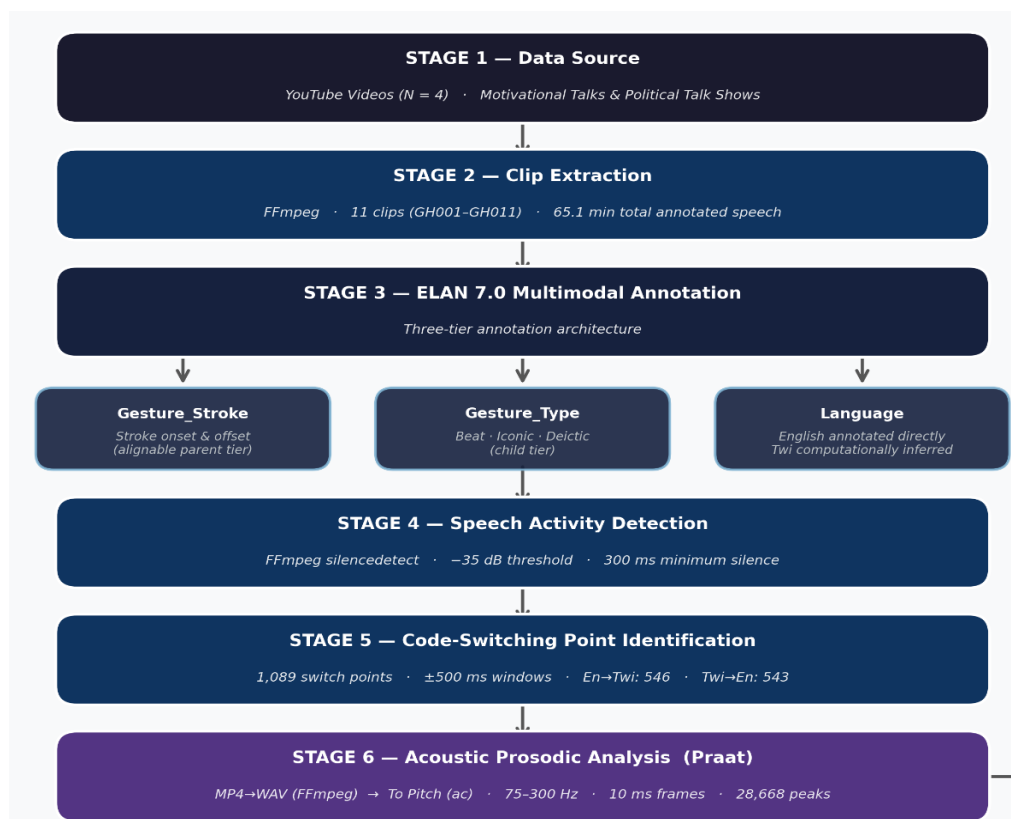
It is important to be transparent about what this analysis captures. Because Praat extracts pitch peaks at high temporal density across continuous speech, almost any gesture onset will fall in proximity to some pitch peak. The analysis therefore establishes the degree of temporal co-

occurrence between gesture strokes and local F0 maxima rather than demonstrating alignment with specific prosodically prominent syllables. Syllable-level correspondence would require full orthographic transcription and phoneme-level forced alignment, which represents a natural extension of this work but falls beyond its current scope.

2.6 Statistical Analysis

All statistical analyses were implemented in Python using the NumPy and SciPy libraries. Three sets of analyses addressed the three research objectives. For the first objective, the distribution of gesture types across the corpus was examined through frequency counts and proportional analysis, providing a descriptive profile of gesture use in Twi-English code-switching discourse. For the second objective, the temporal alignment between gesture stroke onsets and pitch peaks was summarized using descriptive statistics including mean, median, and the distribution of absolute time differences, with results examined both overall and by gesture type. For the third objective, gesture density was computed as the number of gesture strokes per second of speech, calculated separately for regions inside and outside the ± 500 ms switch windows. An independent samples t-test was then used to determine whether gesture density differed significantly between switch-proximal and switch-distal speech, with each of the eleven clips contributing one observation per condition.

Figure 2. Analytical pipeline for the multimodal corpus study of Twi-English code-switching.



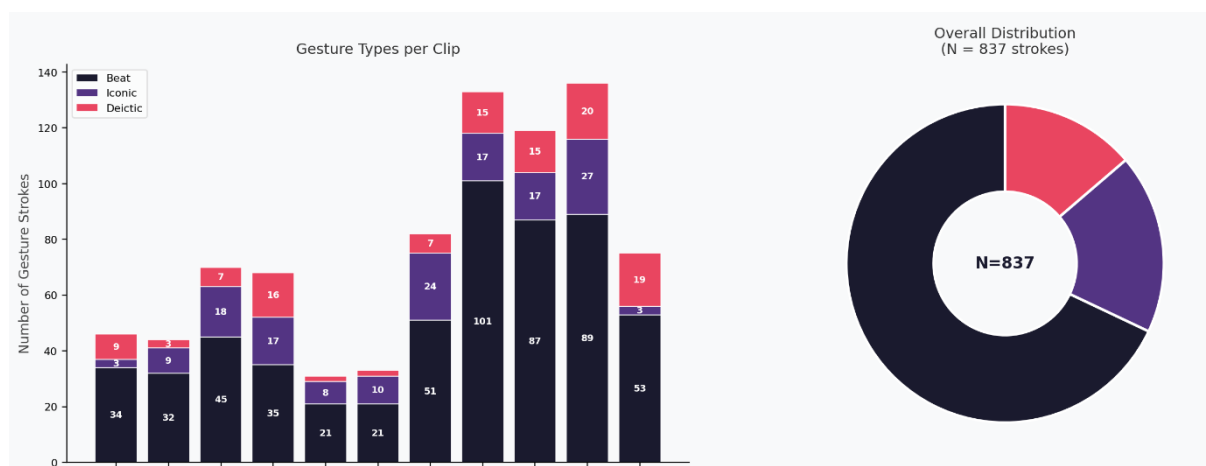
3. Results

This section presents the findings from the analysis of 837 gesture strokes annotated across 65.1 minutes of naturalistic Twi-English speech. Results are organised around the three research objectives: the distribution of gesture types in the corpus, the temporal alignment between gesture strokes and prosodic prominence, and gestural activity at code-switching boundaries.

3.1 Gesture Type Distribution

A total of 837 gesture strokes were annotated across the eleven clips, spanning approximately 65.1 minutes of naturalistic Twi-English bilingual speech. The distribution across McNeill's (2005) three gesture categories was markedly uneven. Beat gestures accounted for the large majority of strokes, with 569 instances representing 68.0% of the total corpus. Iconic gestures followed at 153 strokes (18.3%), and deictic gestures were the least frequent at 115 strokes (13.7%).

Figure 3. *Gesture type distribution, shown as a stacked bar chart per clip on the left and an overall donut chart on the right.*



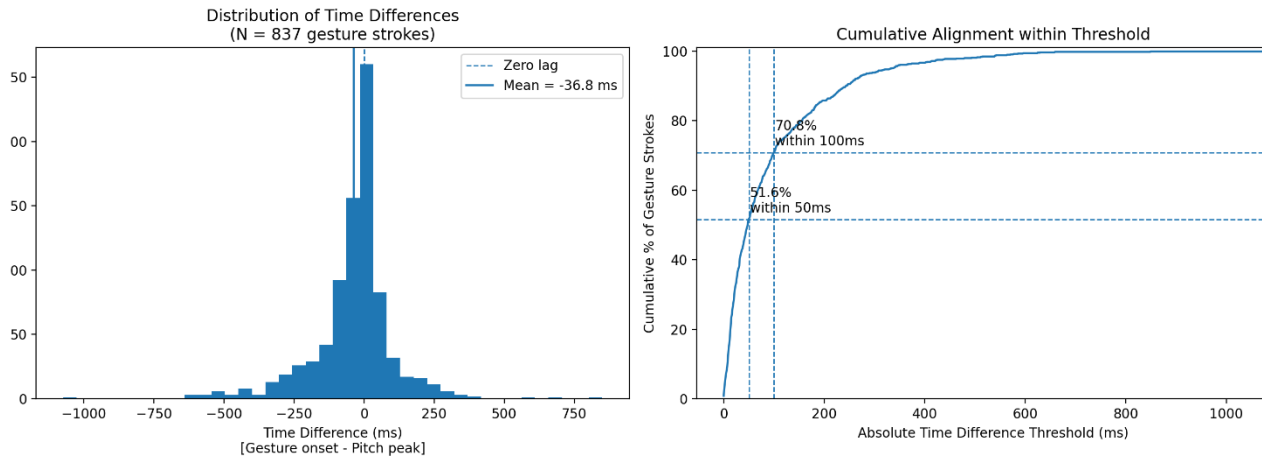
This dominance of beat gestures was consistent across all eleven clips, regardless of speaker or genre. Even in clips with relatively few total strokes, such as GH005 ($n = 31$) and GH006 ($n = 33$), beats constituted the majority. The most gesture-dense clips were GH010 ($n = 136$) and GH008 ($n = 133$), both of which also maintained the same general beat-dominant pattern. Iconic gestures appeared most frequently in clips from Lawyer Anokye Frimpong and Okatakyie Afrifa, where speakers engaged in more descriptive and narrative speech, while deictic gestures were distributed fairly evenly across speakers. The consistency of this pattern across speakers and genres suggests that beat gesture dominance is not an artifact of any single speaker's style but reflects a broader feature of expressive Twi-English code-switching discourse.

3.2 Gesture-Prosody Alignment

To examine the temporal relationship between gesture and prosody, each gesture stroke onset was matched to the nearest local F0 maximum within the same clip, extracted via Praat's

autocorrelation algorithm. The resulting time differences, computed as gesture onset minus pitch peak time in milliseconds, were analyzed both overall and broken down by gesture type.

Figure 4. *Gesture-prosody alignment, shown as a time difference histogram on the left and a cumulative alignment curve on the right, with the 50ms and 100ms thresholds annotated.*



Across all 837 strokes, the mean signed time difference was -36.8ms and the median was -14.0ms, indicating that gesture strokes tended to precede nearby pitch peaks by a small margin. The mean absolute time difference was 93.6ms, with a median of 48.0ms. In terms of coverage, 51.6% of all gesture strokes fell within 50ms of the nearest pitch peak and 70.8% fell within 100ms. Observed differences ranged from -1073ms to +850ms, with most values concentrated close to zero.

Table 2. *Gesture-Prosody Temporal Alignment by Gesture Type*

Gesture Type	N	Mean $ \Delta t $ (ms)	Median $ \Delta t $ (ms)	Within 50 ms (%)	Within 100 ms (%)
Beat	569	97.1	50.0	50.1%	69.8%
Iconic	153	80.9	43.0	55.6%	74.5%
Deictic	115	93.0	46.0	53.9%	71.3%
All	837	93.6	48.0	51.6%	70.8%

Note. $|\Delta t|$ = absolute time difference between gesture stroke onset and nearest pitch peak. Percentages indicate the proportion of gesture strokes falling within the specified threshold.

Alignment was broadly similar across all three gesture types. Beat gestures showed a mean absolute difference of 97.1ms, with 50.1% of strokes within 50ms and 69.8% within 100ms. Iconic gestures were tighter overall at 80.9ms mean absolute difference, with 55.6% within 50ms and 74.5% within 100ms. Deictic gestures showed a mean absolute difference of 93.0ms, with 53.9% within 50ms and 71.3% within 100ms.

It is worth noting that this analysis establishes temporal co-occurrence between gesture strokes and local F0 maxima rather than alignment with specific prosodically prominent syllables, as the latter would require full phonemic transcription and forced alignment. Within the scope of this corpus-based study, the results provide strong evidence that gesture stroke onsets cluster tightly around moments of acoustic prominence in Twi-English bilingual speech.

3.3 Gesture Frequency and Prosodic Prominence at Code-Switching Points

The third analysis examined whether gesture strokes occurred at higher rates in the temporal vicinity of code-switching points. Gesture density was computed as the number of strokes per second of speech, calculated separately for regions inside and outside ± 500 ms windows centered on each of the 1,089 identified switch points.

Figure 5. *Gesture density inside and outside code-switch windows.*

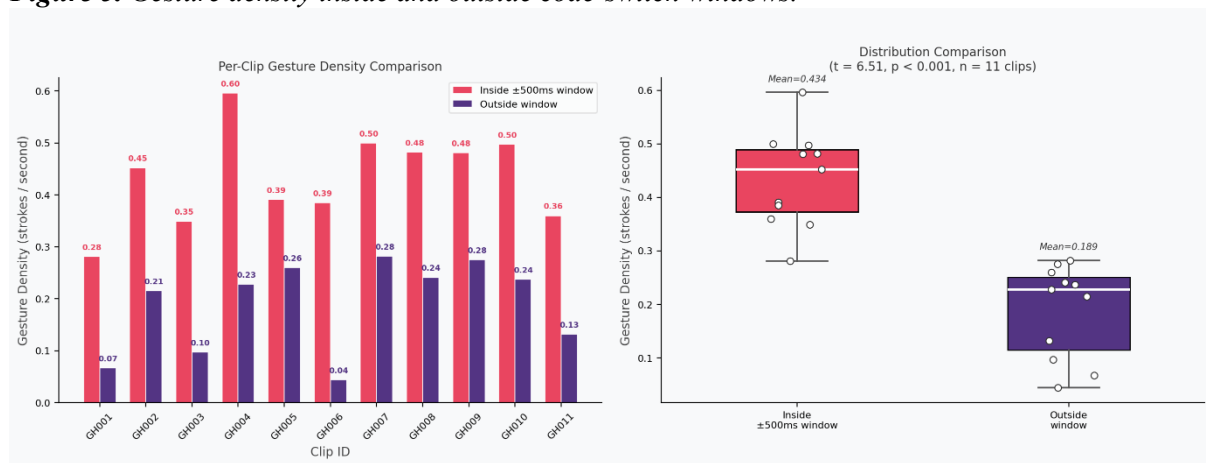


Table 3. *Per-Clip Gesture Density Inside and Outside Code-Switching Windows (± 500 ms)*

Clip	Gestures Inside	Gestures Outside	Density Inside (g/s)	Density Outside (g/s)	Ratio (Inside / Outside)
GH001	32	14	0.281	0.067	4.19×
GH002	28	16	0.452	0.215	2.10×
GH003	46	24	0.349	0.097	3.60×
GH004	31	37	0.596	0.228	2.61×
GH005	18	13	0.391	0.260	1.50×
GH006	30	3	0.385	0.044	8.75×
GH007	34	48	0.500	0.282	1.77×
GH008	79	54	0.482	0.241	2.00×
GH009	39	80	0.481	0.275	1.75×
GH010	76	60	0.497	0.237	2.10×
GH011	50	25	0.360	0.132	2.73×
Mean			0.434	0.189	2.30×

Note. Density values represent gesture strokes per second of speech. Ratio = density inside / density outside. Independent samples t-test: $t(10) = 6.51, p < .001$.

The results show a clear and consistent elevation of gestural activity at code-switching boundaries. Mean gesture density inside the switch windows was 0.434 strokes per second, compared to 0.189 strokes per second outside, representing a ratio of approximately 2.30 to 1. Across all eleven clips, gesture density was higher inside the switch windows than outside, with per-clip ratios ranging from 1.50 in GH005 to 8.75 in GH006. The one notable outlier is GH007, where the raw gesture count was higher outside the windows ($n = 48$) than inside ($n = 34$), yet even here the density ratio favoured the switch-proximal region at 1.77 to 1, reflecting the larger total speech time outside the windows.

An independent samples t-test confirmed that the difference in gesture density between switch-proximal and switch-distal speech was statistically significant, $t(10) = 6.51, p < .001$. The effect was robust across speakers and genres, appearing in motivational address clips as well as political talk show segments, which strengthens confidence that the finding reflects a genuine multimodal feature of code-switching behavior rather than a speaker-specific or genre-specific tendency.

4. Discussion

This study examined how gesture and prosody work together in naturalistic Twi-English bilingual speech. Three questions guided the analysis: what types of gestures speakers produce, how closely gesture strokes align with pitch peaks in the speech signal, and whether gestural activity increases around code-switching points. The results across all three questions point in the same direction. Gesture in this corpus is not incidental to speech. It is systematically tied to it, both at the level of individual prosodic peaks and at the broader level of language transitions.

The distribution of gesture types is a sensible starting point. Beat gestures made up 68.0% of the 837 annotated strokes, which makes sense when you consider the nature of the speech in this corpus. Political commentary and motivational oratory are genres that demand a lot from speakers. They need to hold attention, signal which parts of their message carry the most weight, and keep the rhythm of their argument going. Beat gestures serve exactly that function. They carry no visual meaning in themselves. They are, essentially, the manual equivalent of spoken stress, marking emphasis and prominence through repeated rhythmic movement (McNeill, 2005). Research across a range of languages and speech styles has consistently found that beat gestures dominate in formal, speaker-driven discourse, precisely because they are the primary gestural tool for prosodic and discourse structuring (Loehr, 2012; Brown & Prieto, 2021). Mendoza-Denton and Jannedy (2011) documented this pattern specifically in political speech, showing that gesture and intonation layer together to construct prominence and stance in public discourse contexts. Shattuck-Hufnagel and Ren (2018) extended this observation to academic lecture-style speech, finding that non-referential gesture strokes consistently aligned with pitch-accented syllables across 1,334 gestures, reinforcing the view that beat gesture production is organized around prosodic structure rather than semantic content. Iconic and

deictic gestures appeared more selectively, clustering around moments of description and direct reference, as would be expected. This pattern is also consistent with what Gullberg (2013) reports for bilingual speakers. In her review of bilingual adult gesture production, she notes that gestures serving pragmatic and discourse-structuring functions are among the most frequent in bilingual speech, appearing precisely where the demands of managing complex, fluent output require manual rhythmic support (p. 425). Beat gestures fall squarely into that category. The fact that the same pattern dominates in Twi-English code-switching discourse suggests that the prosodic structuring role of beat gestures is not weakened by the demands of bilingual management. If anything, it may be strengthened by them. This reading is supported by Gullberg's (1998, pp. 76-77) earlier empirical work, which found that bilinguals produced more gestures in their second language overall, and that the category showing the sharpest increase was beats, the gesture type most directly tied to prosodic and discourse organisation.

The median time difference between gesture stroke onset and the nearest pitch peak was -14.0ms. 51.6% of all strokes fell within 50ms of a local F0 maximum and 70.8% fell within 100ms. To appreciate what that means practically, 50ms is roughly the length of a single speech sound. Gesture strokes are not loosely associated with moments of prosodic prominence. They land on them, consistently and precisely, across all speakers and genres in the corpus. Pouw, Trujillo and Dixon (2020) confirm this level of synchrony across both device-based and video-based data collection methods, validating its consistency as a measurable phenomenon. Pouw and Dixon (2019) showed that gesture kinematic peaks and F0 peaks entrain to each other even when auditory feedback is disrupted, and Pouw, Harrison and Dixon (2019) went further by providing biomechanical evidence that the physical effort of producing a gesture directly influences the acoustic properties of the accompanying speech, including pitch. That finding raises the possibility that the coupling observed here is not purely cognitive. It may be partly mechanical, built into the physical act of speaking and gesturing at the same time. Esteve-Gibert and Prieto (2013) documented similar alignment in controlled Catalan speech tasks, while Loehr (2012) found it in naturalistic English conversation. The current study adds naturalistic bilingual Twi-English discourse to that growing body of evidence.

What stands out further is that the alignment pattern was broadly similar across all three gesture types. Beat, iconic, and deictic strokes showed mean absolute differences in the 80–97 ms range (Beat 97.1 ms, Iconic 80.9 ms, Deictic 93.0 ms), with no meaningful difference between them. That uniformity is worth pausing on. A rhythmic coupling between beat gestures and pitch peaks might be expected, since both are fundamentally about marking rhythmic prominence. But iconic and deictic gestures are not rhythmic in the same way. They carry visual and referential content. The fact that they align just as tightly suggests that the coordination runs deeper than simple rhythm matching. The whole gesture-speech system appears to be organized around prosodic peaks, regardless of what any individual gesture is communicating. McNeill and Duncan (2000) proposed exactly this kind of unified architecture in their growth point framework, arguing that gesture and speech are not two parallel systems but a single cognitive unit. The stroke and the pitch peak converge because they are both expressions of the same underlying communicative moment. The results here are consistent with that view. It is also worth noting that this pattern holds in a bilingual corpus where speakers are continuously

managing two phonological systems. Graziano, Nicoladis and Marentette (2020) found no significant difference in gesture-speech alignment between monolingual and bilingual speakers in an English-French study, and the current findings suggest the same holds for Twi-English bilingualism. Language switching, it seems, does not break the tight coupling between gesture and speech.

One important qualification needs stating clearly. Each gesture stroke onset was matched to the nearest F0 peak in the Praat pitch contour. Because Praat extracts pitch at 10ms intervals across all voiced speech, there are a lot of peaks in the signal, and any gesture onset will naturally land near one. What the analysis demonstrates is that gesture strokes cluster around moments of local acoustic prominence rather than at arbitrary positions in the speech stream, which is a meaningful and replicable finding. What it does not demonstrate is that speakers are precisely targeting specific prosodically accented syllables. That level of detail would require full phonemic transcription and forced alignment, which is a clear direction for future work.

One theoretical tension deserves direct acknowledgement. The introduction argued that pitch functions as a lexical cue in Twi but an intonational cue in English, yet the acoustic analysis treats all F0 maxima identically regardless of which language is active at that moment. The alignment findings therefore cannot specify whether gesture strokes are coupling with lexical tone peaks, intonational prominence peaks, or both. What they do establish is that coupling holds consistently across a bilingual signal where pitch is simultaneously serving two phonological functions, a result that suggests the biomechanical basis for gesture-speech synchrony described by Pouw, Harrison and Dixon (2019) may operate independently of the functional status of the peak driving it. Whether alignment precision differs between Twi and English segments is a question that requires language-tagged F0 analysis and forced alignment to answer, and it represents a clear direction for future work.

The third finding is the most novel. Gesture density inside the ± 500 ms windows around code-switching points was 2.30 times higher than outside those windows, a difference that was statistically significant at $t(10) = 6.51$, $p < .001$. To understand why this is interesting, it helps to think about what a code-switching point actually involves for a speaker. Switching between Twi and English is not just a vocabulary decision. It requires real-time adjustment of phonology, rhythm, and often the speaker's stance toward what they are saying. Torres Cacoullos (2020) has described code-switching boundaries as sites of prosodic recalibration, where speakers use intonational and rhythmic cues to manage the transition between language systems. Gullberg (2013) adds a gesture-specific dimension to this picture. Documenting language contact communities in which spoken language shift is underway, Gullberg reports that gestural behaviour at language boundaries changes more slowly than the spoken language itself. There is an intermediate stage in which both gestural repertoires coexist and gesture shift lags language shift rather than accompanying it in lockstep (p. 428). This is relevant to the present finding because it suggests that gestural activity at code-switching transitions is not simply a reflex of the spoken switch. It has its own organizational logic that responds to the demands of the boundary itself. The present results suggest that this boundary work extends beyond the acoustic channel. Speakers are also marking the switch with their hands. That

finding fits with broader evidence that gesture rates increase at moments of discourse salience and structural transition (Brown & Prieto, 2021; Jiménez-Bravo & Marrero-Aguiar, 2024). What this study contributes specifically is showing that code-switching boundaries are one such site of multimodal amplification in Twi-English bilingual speech. The gesture does not just accompany the switch. It signals it. The fact that elevated density ratios appeared in all eleven clips, ranging from 1.50 to 8.75 across clips, adds confidence that this is a genuine and consistent feature of how these speakers communicate rather than a pattern driven by one or two recordings.

This study has limitations worth acknowledging. The corpus covers public-facing speech only, which limits generalizability to everyday conversational code-switching. Gesture annotation was carried out by a single coder without inter-rater verification. The language timeline was built through computational inference, meaning non-linguistic vocalizations are absorbed into Twi intervals and may introduce some noise into switch point detection. The t-test also rests on eleven clips, which is a modest sample for inferential testing.

One theoretical clarification deserves explicit statement. This study does not test whether Twi gestures differ formally from English gestures, nor whether speakers adopt different gestural styles or repertoires in their two languages. The prior literature on gesture and code-switching, specifically Lacroix and Rioux (1978), who found no evidence of gestural behaviour switching when language switched, and Newbury (2011), who documented emblem change at the community level rather than the moment-to-moment level addressed questions of gestural form and repertoire. The present findings do not contradict those results. They operate at a different level of analysis entirely: the question here is whether gesture density and temporal precision change at the structural boundary of a language switch, regardless of what kind of gesture is produced. The answer is yes. Code-switching boundaries are moments of heightened gestural activity, and that elevation is consistent across speakers and genres. Prior studies that found no form-level switching did not examine this dimension, and the present results should be read as supplementing that body of work rather than revising it.

5. Conclusion

This study set out to examine whether gesture and prosody work together systematically in Twi-English bilingual speech, and the findings confirm that they do. Gesture strokes cluster near local pitch peaks with a median time difference of -14.0ms, with broadly similar patterns across beat, iconic, and deictic gesture types. Gesture density is 2.30 times higher in the moments surrounding code-switching points than in the surrounding speech, a difference that is statistically robust and consistent across all eleven clips and all four speakers. Beat gestures dominate the corpus at 68.0%, reflecting the prosodic and discourse-structuring demands of the genres studied.

What these findings mean, taken together, is that code-switching in Twi-English bilingual discourse is not a purely spoken event. It is multimodal. When speakers transition between languages, their hands mark that transition just as their voices do. Gesture and prosody are not running in parallel channels that occasionally line up. They are functioning as a unified

communicative system, organized around the same moments of prominence and the same structural boundaries in the speech stream. That finding matters because it challenges the default assumption in much of the code-switching literature that language alternation is primarily a linguistic phenomenon best studied through transcripts and acoustic data alone.

It also matters for what it suggests about bilingual communicative competence more broadly. Managing two languages in real time is cognitively demanding, yet the gesture-speech coupling observed here is as tight as anything reported for monolingual speakers in comparable studies. Bilingual speakers are not simply toggling between two spoken systems. They are coordinating an integrated multimodal system while doing so, and they appear to do it fluently.

This observation connects to two foundational frameworks in bilingualism research. Grosjean (1998) argued that bilinguals are not simply two monolinguals sharing a single mind. They constitute a third type of speaker whose communicative system needs to be understood on its own terms. Cook (2002) built on this through the concept of multicompetence, the idea that knowing more than one language fundamentally changes the mind. Not because it installs two separate systems, but because it develops a qualitatively different kind of competence that encompasses both. The multimodal findings of the present study extend both frameworks into the gestural domain. The gesture-speech coupling observed here is as precise and consistent as anything reported for monolingual speakers, and it holds even as speakers navigate two prosodically distinct language systems in real time. This suggests that bilingual multimodal competence is not a reduced or compromised version of monolingual competence. It is a different form of it, and in some respects a more demanding one, maintaining gestural precision while managing language alternation simultaneously. Gullberg (2013) observed that bilinguals in her review looked like themselves across both modalities and languages (p. 428). The Twi-English speakers in this corpus are no different. They are not approximating monolingual gestural behaviour in either language. They are doing something that monolingual frameworks were simply not built to describe.

For the field of multimodal data science, this study demonstrates the value of a fully computational, reproducible pipeline for bilingual corpus analysis, combining ELAN annotation, automated speech detection, Praat pitch extraction, and Python-based statistical analysis into an end-to-end workflow that can be adapted and extended by other researchers. Future work should build on this foundation with a larger and more genre-diverse corpus, a second annotator, explicit coding of non-linguistic vocalizations, and a forced alignment pipeline that allows gesture-prosody mapping at the syllable level. Extending this approach to other African language contact settings would also help establish whether the patterns observed here are specific to Twi-English bilingualism or reflect a more general feature of how multimodal systems behave at language boundaries.

Twi-English bilingual speakers use their bodies and their voices together, and they do so with a precision and consistency that reflects deep integration between the gestural and prosodic channels of human communication. That is the central finding of this study, and it opens a line of inquiry into the multimodal dimensions of African bilingual discourse that has barely begun.

AI Disclosure

The author used Grammarly AI to assist with proofreading, sentence restructuring and grammar checking during manuscript preparation. All intellectual content analysis, interpretation and conclusions are the authors own.

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